NLSE：

$$i\frac{\partial A}{\partial z}+\frac{i\alpha}{2}A-\frac{\beta_2}{2}\frac{\partial^2 A}{\partial T^2}+\gamma|A|^2A=0.$$

令 $A=\sqrt{P_0}\,e^{-\alpha z}|u|$,

$$i\frac{\partial U}{\partial z}-\frac{\beta_2}{2}\frac{\partial^2 U}{\partial z^2}+\gamma P_0 e^{-\alpha z}|u|^2U=0.$$

矩方法

假设 脉冲光纤传输时，能保持自己的形状，

基本思想：

将脉冲处理为能量，均方根宽度，啁啾分别为 E_p，σ_p，C_p 的粒子.

$$E_p=\int_{-\infty}^{\infty}|u|^2\,dT,$$

$$\sigma_p^2=\frac{1}{E_p}\int_{-\infty}^{\infty}T^2|u|^2\,dT.$$

$$C_p=\frac{i}{E_p}\int_{-\infty}^{\infty}T\left(u^*\frac{\partial u}{\partial T}-u\frac{\partial u^*}{\partial T}\right)dT.$$

三参量均随脉冲传输变化，如何变化？对 z 取微分：

$$\begin{cases}\dfrac{dE_p}{dz}=0\\[2ex]\dfrac{d\sigma_p^2}{dz}=\dfrac{\beta_2}{E_p}\displaystyle\int_{-\infty}^{+\infty}T^2\,\mathrm{Im}\left(u^*\frac{\partial^2 u}{\partial T^2}\right)dT\\[2ex]\dfrac{dC_p}{dz}=\dfrac{2\beta_2}{E_p}\displaystyle\int_{-\infty}^{\infty}\left|\frac{\partial u}{\partial T}\right|^2dT+e^{-\alpha z}\frac{\gamma P_0}{E_p}\int_{-\infty}^{\infty}|u|^4\,dT.\end{cases}$$

For chirp Gaussian pulse:

$$U(z,T) = a_p \exp\left[-\frac{1}{2}(1+iC_p)(T/T_p)^2 + i\phi_p\right]$$

- a_p, C_p, T_p, ϕ_p 为 z 函数.
- $E_p = \sqrt{\pi}\, a_p^2 T_p$ 不随 z 变化，可用 $E_0 = \sqrt{\pi}\, T_0$ 代替.
- $T_p = \sqrt{2}\,\sigma_p$

代入：

$$\begin{cases} \dfrac{dT_p}{dz} = \dfrac{\beta_2 C_p}{T_p} \\ \dfrac{dC_p}{dz} = (1+C_p^2)\dfrac{\beta_2}{T_p^2} + \dfrac{\gamma P_0 T_0}{\sqrt{2}\,T_p} e^{-\alpha z} \end{cases} \Rightarrow T_p^2(z) = T_0^2 + 2\int_0^z \beta_2(z) C_p(z)\, dz$$

变分法.

拉格朗日函数

$$L \equiv \int_{-\infty}^{\infty} L_d(q, q^*)\, dT.$$

L_d: 拉格朗日密度.

要使作用量

$$S \equiv \int L(z)\, dz$$

最小, 要满足欧拉 - 拉格朗日方程:

$$\frac{\partial}{\partial T}\left(\frac{\partial L_d}{\partial q_T}\right) + \frac{\partial}{\partial z}\left(\frac{\partial L_d}{\partial q_z}\right) - \frac{\partial L_d}{\partial q} = 0.$$

$q_T \equiv \frac{\partial q}{\partial T}$, $q_z \equiv \frac{\partial q}{\partial z}$

变分法: NLSE 可由拉格朗日密度导出:

$$L_d = \frac{i}{2}\left(U^* \frac{\partial U}{\partial z} - U \frac{\partial U^*}{\partial z}\right) + \frac{\beta_2}{2}\left|\frac{\partial U}{\partial T}\right|^2 + \frac{1}{2}\gamma P_0 e^{-\alpha z}|U|^4.$$

其中, U^* 相当于广义坐标 q,

对由前文给出的高斯脉冲:

$$U(z,T) = a_p \exp\left[-\frac{1}{2}(1+iC_p)(T/T_p)^2 + i\phi_p\right]$$

$$L = \frac{\beta_2 E_p}{4T_p^2}(1+C_p^2) + \frac{\gamma e^{-\alpha z} E_p^2}{\sqrt{\pi}\, T_p} + \frac{E_p}{4}\left(\frac{dC_p}{dz} - \frac{2C_p}{T_p}\frac{dT_p}{dz}\right) - E_p \frac{d\phi_p}{dz}.$$

其中, $E_p \equiv \sqrt{\pi} a_p^2 T_p$ 为脉冲能量.

代入欧拉-拉格朗日方程：

$$\frac{d}{dz}\left(\frac{\partial L}{\partial g_z}\right)-\frac{\partial L}{\partial g}=0$$

其中，$g_z=\frac{dg}{dz}$，g表示4个脉冲参量，E_p、T_p、c_p、ϕ_p.

由 $g=\phi_p$，$\Rightarrow$ $\frac{dE_p}{dz}=0$.

由 $g=E_p$，$\Rightarrow$ $\frac{d\phi_p}{dz}=\frac{\beta_2}{2T_p^2}+\frac{5\gamma e^{-\alpha z}E_p}{4\sqrt{2\pi}\,T_p}$.